

G. Lejeune Dirichlet

Werke, Band II, XXXIX

Remark on Dirichlet's Works, vol. I, p. 348, line 7

By R. Dedekind; English translation

If the prime $p \equiv 5 \pmod{8}$, then the two numbers h and k satisfying

$$h^2 - pk^2 = -4$$

are not always odd, as is asserted there; they can also both be even. The latter case first occurs for $p = 37$, where $h = -12$ and $k = 2$. The method of solving Pell's equation by means of cyclotomy arose from the investigations on the arithmetic progression and on the class number of binary quadratic forms of positive determinant; and that assertion would have found its correction of itself in the memoir *Recherches sur diverses applications de l'analyse infinitesimale a la theorie des nombres*, if Dirichlet had there returned to the consideration of the individual cases. The lectures he gave in Göttingen in the winter 1856–1857 likewise broke off before this point was settled; only while preparing them for publication did I notice the error just mentioned.

Let T, U, T', U', T'', U'' denote the least natural numbers satisfying

$$T^2 - pU^2 = 1, \quad T'^2 - pU'^2 = 4, \quad T''^2 - pU''^2 = -4.$$

Putting the fundamental unit in the form

$$\frac{T'' + U''\sqrt{p}}{2} = \varepsilon,$$

one sees at once that

$$\frac{T' + U'\sqrt{p}}{2} = \varepsilon^2, \quad T' \equiv U' \equiv T'' \equiv U'' \pmod{2}.$$

If one further sets

$$T + U\sqrt{p} = \varepsilon^{2m},$$

Dirichlet shows (vol. I, pp. 470–471) that $m = 1$ or $m = 3$, according as the numbers T', U' , and hence also T'', U'' , are even or odd. If n and n' denote the class numbers, there denoted by h and h' , of the forms of the first and second kind belonging to determinant p , then at the same time

$$mn = 3n'.$$

By §11 of the same memoir (vol. I, p. 495), where one has to put $P = p$, $Y_1 = q = pk$, and $Z_1 = -h$, the class number n is determined by

$$(T + U\sqrt{p})^n = \left(\frac{-h + k\sqrt{p}}{2} \right)^6.$$

Since the numbers $-h$ and k are always positive, it follows that

$$\frac{-h + k\sqrt{p}}{2} = \varepsilon^{n'}.$$

Consequently h, k will certainly be even when T'', U'' , hence also T', U' , are even, that is, when $m = 1$, and therefore $n = 3n'$. In the *Disquisitiones Arithmeticae*, Art. 256 VI, a passage cited by Dirichlet at the end of §8 for comparison, Gauss notes that this case $n = 3n'$ occurs for

$$37, 101, 197, 269, 349, 373, 389, 557$$

and for no other prime below 600. For example,

$$-h = 12, \quad k = 2, \quad n' = 1, \quad n = 3 \quad (p = 37),$$

$$-h = 20, \quad k = 2, \quad n' = 1, \quad n = 3 \quad (p = 101).$$

The preceding equation for the class number n immediately yields the general criterion that h, k are even always and only when n is divisible by 3. This certainly happens in the case $m = 1$, because then $n = 3n'$; but it can also happen in the other case $m = 3$, $n = n'$, although T', U', T'', U'' are odd. The smallest prime p for which this phenomenon occurs is 229; the extremely laborious computation of the functions Y, Z gives

$$-h = 3420, \quad k = 226 \quad (p = 229).$$

The same result is obtained much more quickly by setting up the reduced forms, from which

$$T'' = 15, \quad U'' = 1, \quad n = n' = 3$$

follows.

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